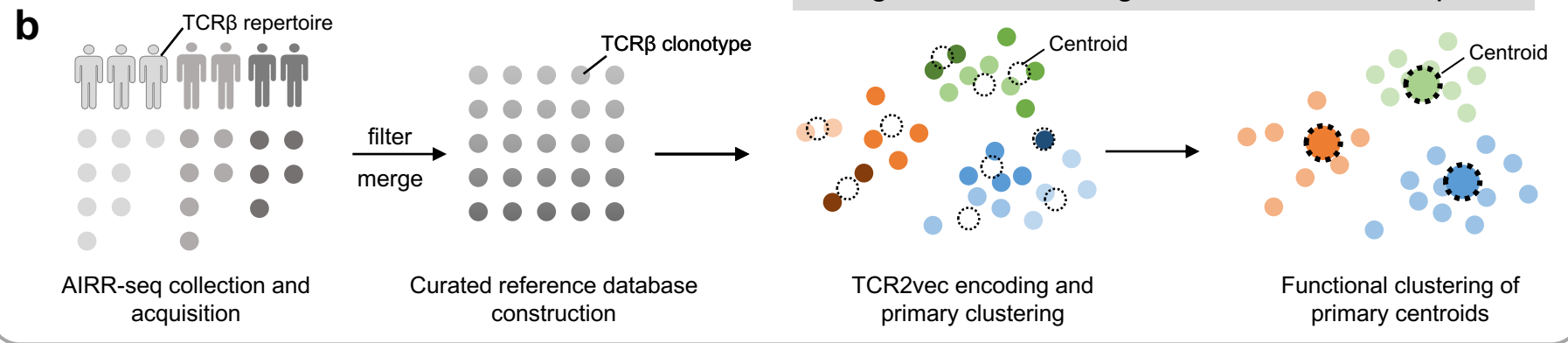


Stage 1: Constructing the reference TCR space



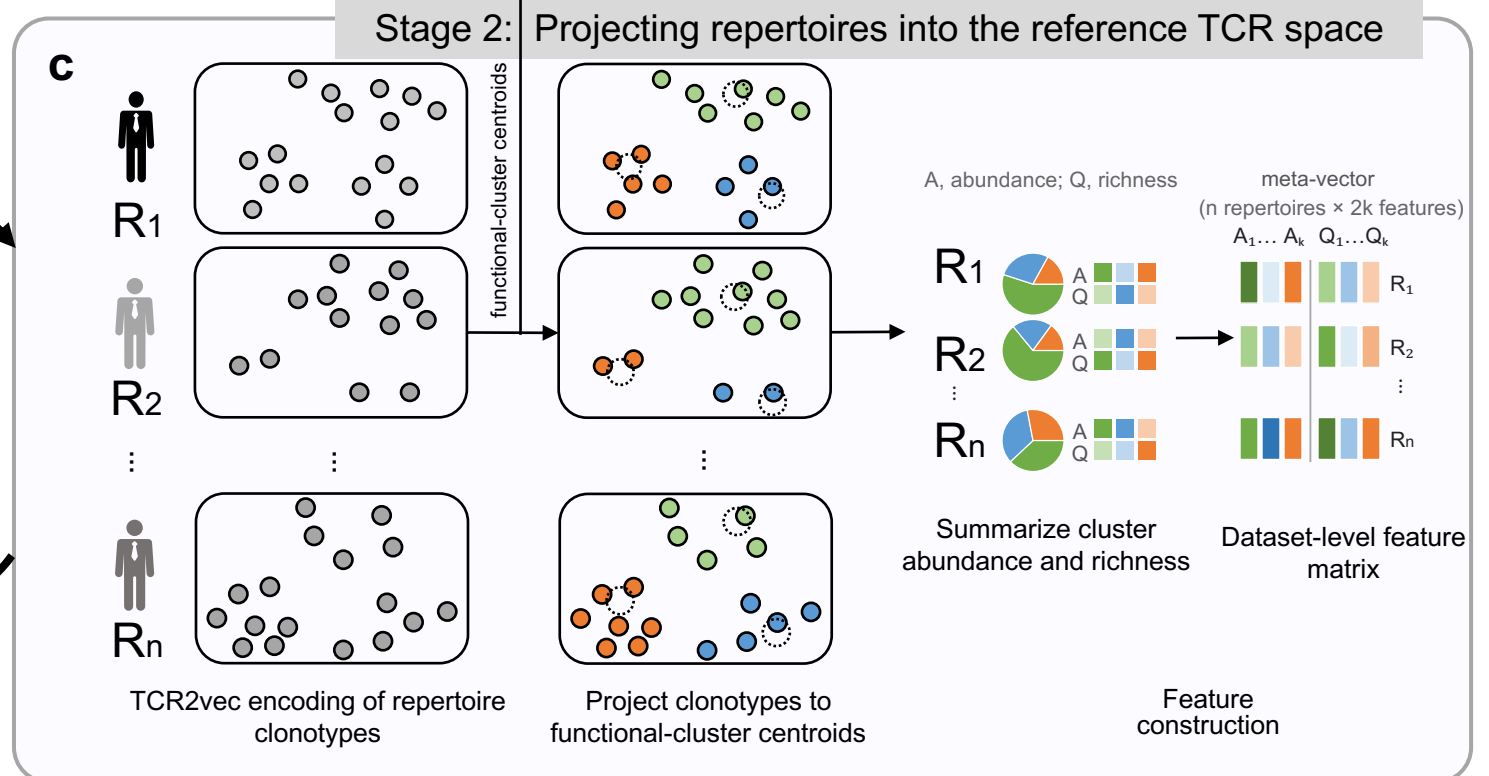
Stage 2: Projecting repertoires into the reference TCR space

a

Input: bulk TCR β repertoires

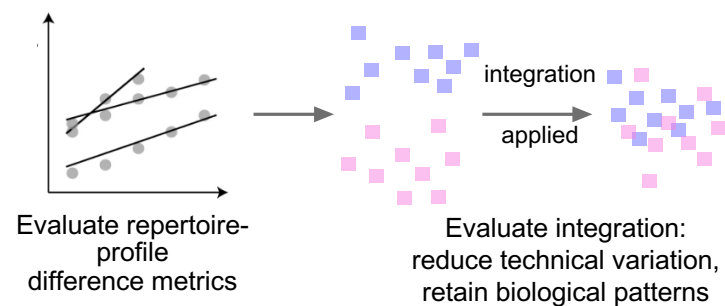
c

Output: repertoire meta-vectors



Stage 3: Framework evaluation with simulated and real-world data

d



Stage 4: Downstream analysis of repertoire profiles

e

